

The Deskinator

Section: A8 Group: 1

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1. Executive Summary

The Deskinator is a fully autonomous desktop-cleaning robot designed to clear a standard 4 ft × 4 ft office desk within 2 minutes without any user intervention after activation. The robot uses an S-shaped lawnmower navigation strategy to ensure broad surface coverage. It also uses a custom-built vacuum system consisting of a motorized fan and nozzle to collect debris into a removable container. Edge detection is handled by HC-SR04 ultrasonic sensors to prevent the robot from falling off the desk. The Deskinator uses two separate power sources: a 9 V battery to power the vacuum fan and suction system, and a 12 V battery to power the Arduino, sensors, motor driver, and drive motors, keeping the total system weight under 750 g. Upon completion, the robot signals the user via an LED and halts all motion and suction. Testing confirmed a 70% overall pass rate (14/20 trials), demonstrating that the final design was able to meet the major functional requirements of the project.

2. Introduction, Problem Statement, Objectives, and Metrics

The Deskinator is a touchless, battery-powered autonomous robot designed to clean a standard 4×4 ft desk surface by removing visible debris while safely navigating the desktop environment. Because the robot operates on an elevated surface, preventing falls while maintaining effective cleaning coverage is a central design challenge.

Key Objectives and Metrics:

- Achieve at least 90% desk coverage within 2 minutes
- Operate fully on battery power for at least 5 minutes per charge
- Reliably detect desk edges and prevent falls during operation
- Maintain a total mass below 750 g
- Start the cleaning cycle without direct physical contact
- Collect both paper scraps and rice using a student-built vacuum system
- Remain within the \$200 project budget

3. Important Design Features and Basis for Selection

Several key design decisions were made during the development of the Deskinator. The most important choices and their rationale are described below.

Navigation Strategy: The Deskinator uses an S-shaped lawnmower navigation strategy. After activation, the robot moves straight until the front ultrasonic sensor detects the desk edge. It then checks the left and right sensors, reverses slightly, turns left or right based on the available space, and continues straight in the opposite direction. This back-and-forth pattern repeats during the cleaning cycle and helps the robot cover about 90% of the desk surface within 120 seconds.

Drive System: DC motors were chosen over stepper motors because they are simpler, lighter, and consume less power. They also provide sufficient speed and control for desktop navigation.

Vacuum System: The cleaning mechanism uses a custom-built DC fan motor and nozzle to lift debris into a removable container. A brush-only approach was rejected because it could scatter lightweight debris instead of containing it.

Edge Detection: HC-SR04 ultrasonic sensors detect desk edges by measuring sudden increases in distance, which triggers the robot to reverse and reorient. Ultrasonic sensors were chosen because they provide reliable non-contact measurement and are easy to integrate with Arduino.

Touchless Start: The robot is activated by a gesture sensor, reducing the risk of displacing the robot at startup. A clap sensor was considered but rejected because it would be less reliable in noisy environments.

4. Description of Final Product

The Deskinator is a rectangular-chassis differential-drive robot with a footprint of 24×20 cm. The chassis is constructed from wood, keeping the structure lightweight and easy to modify, and the nozzle component is 3D-printed to maximize airflow concentration at the desk surface.

The robot uses two separate battery sources. A 9 V battery powers the vacuum fan for suction, while a 12 V battery powers the Arduino microcontroller, sensors, motor driver, and drive motors. Two DC motors drive the wheels independently for differential steering, and a gesture sensor detects the user's hand motion to initiate the cleaning cycle without physical contact. The vacuum assembly consists of a DC fan motor connected to the 3D-printed nozzle, channeling debris into a removable container. HC-SR04 ultrasonic sensors mounted around the chassis perimeter detect sudden drop-offs and command the robot to reverse and reorient. The total system weight is nearly 1000 g. Upon completing the cleaning cycle, the robot halts all motion and suction and activates an LED to notify the user.

5. Evaluation of Results

After testing, the Deskinator achieved a 70% pass rate (14/20 trials). A trial was counted as a pass when the robot achieved $\geq 90\%$ surface coverage within 120 s and successfully avoided falling off the desk. Trials that experienced edge-detection failure were terminated early, so no coverage or timing data were recorded for those runs. Detailed trial data and plots are provided in Appendix I (Figure I).

Three main failure modes were observed across the failed trials. First, edge-avoidance issues accounted for the majority of failures — the robot occasionally did not decelerate or reverse quickly enough after an edge was detected, suggesting a sensing-to-actuation response delay and a non-zero stopping distance. Additionally, when turning at table corners, the rectangular chassis

can swing outward about the differential-drive rotation center, causing part of the device to overhang the desk edge. Second, in some trials the system stopped even when measured coverage had not exceeded 90%, indicating that the current coverage measurement may be inconsistent. Third, timing failures occurred when the robot encountered edges multiple times, leading to repeated avoidance maneuvers and a longer path length, which can cause the total operation time to exceed the 2-minute limit.

6. Lessons Learned

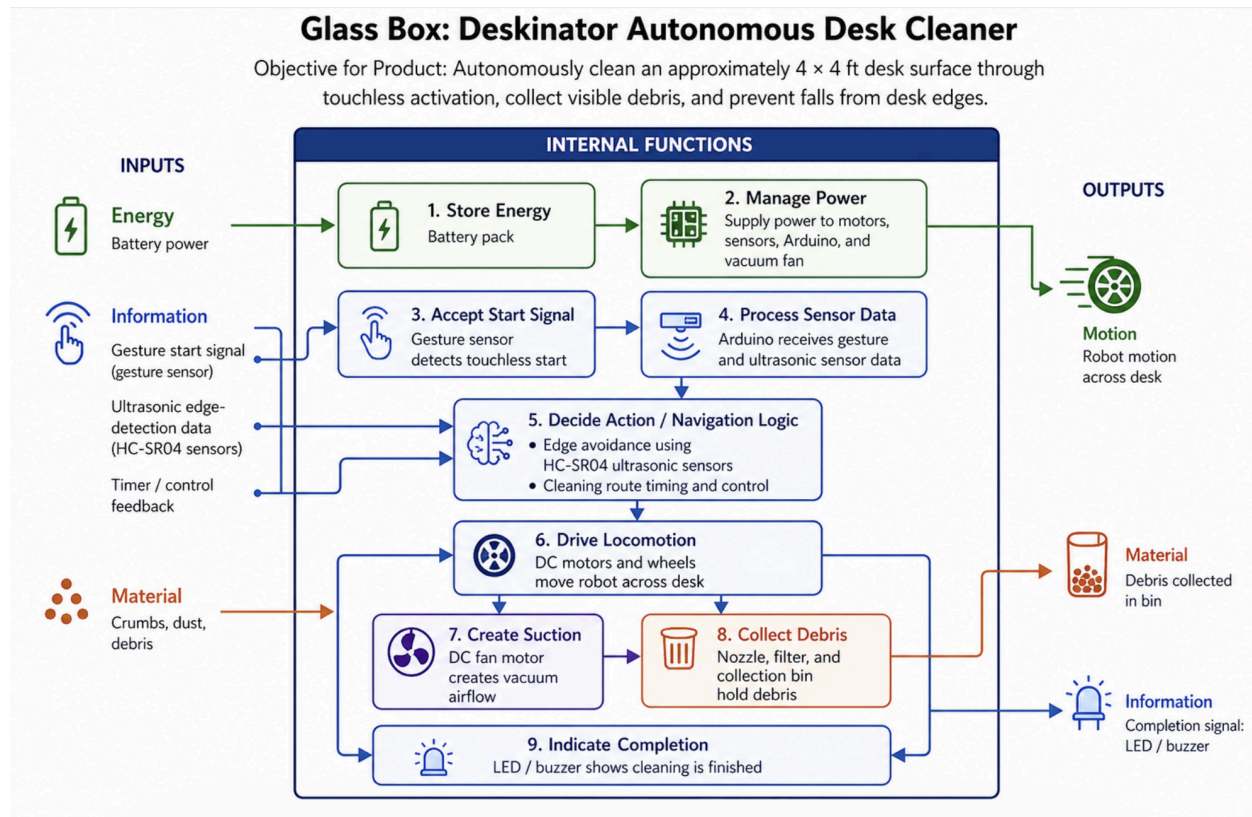
Through this project, continuous trial-and-error and investigation significantly improved our understanding of the overall product design process, especially our knowledge of voltage and circuit principles. For example, we learned how to determine the required current and voltage to support the operation of the entire system. At the same time, we also developed troubleshooting skills. Since unexpected issues frequently occurred during the process, we learned how to systematically identify and resolve problems through these experiences.

7. References

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Appendix

Appendix A — Glass box



Appendix B — PCC (Objectives + Constraints)

Objectives:

Pairwise Comparison Chart For Objectives							
Objectives	Safety / Fall Prevention	Coverage within 2 Minutes ($\geq 90\%$)	Debris Collection (paper + rice)	Wireless Runtime (≥ 5 min)	Compact & Lightweight (< 750 g)	Budget ($\leq \$200$)	Total
Safety / Fall Prevention		1	1	1	1	1	5
Coverage within 2 Minutes ($\geq 90\%$)	0		1	1	1	1	4
Debris Collection (paper + rice)	0	0		1	1	1	3
Wireless Runtime (≥ 5 min)	0	0	0		1	1	2
Compact & Lightweight (< 750 g)	0	0	1	1		1	1
Budget ($\leq \$200$)	0	0	0	0	0		0

Constraints:

Pairwise Comparison Chart For Constraints							
Constraints	Edge Detection Required	Wireless Operation	Runtime ≥ 5 min	Touchless Start	Mass < 750 g	Budget $\leq \$200$	Total
Edge Detection Required		1	1	1	1	1	5
Wireless Operation	0		1	1	1	1	4
Runtime ≥ 5	0	0		1	1	1	3

min							
Touchless Start	0	0	0		1	1	2
Mass < 750 g	0	0	0	0		1	1
Budget ≤ \$200	0	0	0	0	0		0

Appendix C — Morph chart

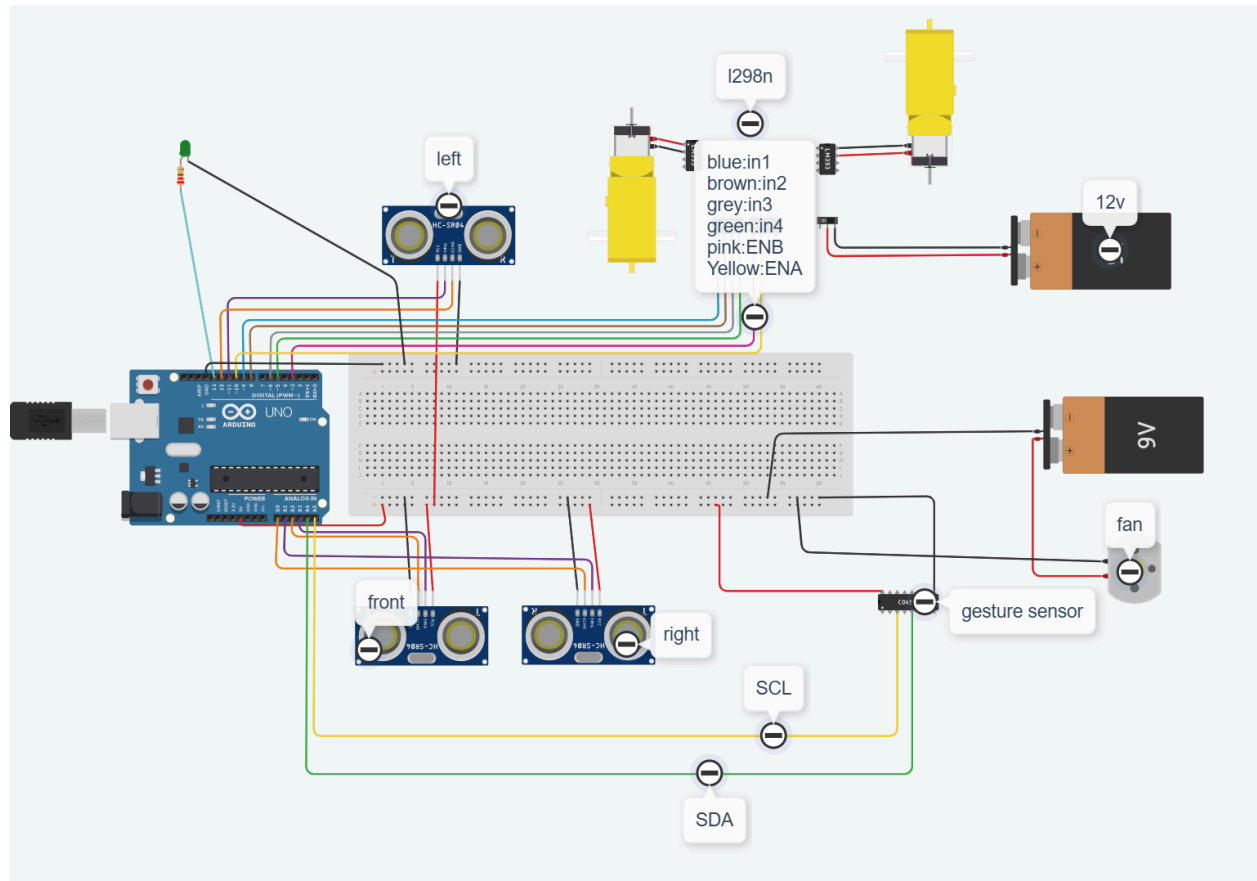
Function (What it must do)	Option A	Option B	Option C	Final Choice (yours)
Touchless Start	Clap sensor (sound)	Gesture sensor (APDS9960)	IR proximity start	Gesture sensor (APDS9960)
Edge Detection	Ultrasonic (HC-SR04)	IR reflective “cliff” sensor	Mechanical bumper switch	Ultrasonic (HC-SR04)
Drive / Locomotion	DC motors (differential drive)	Stepper motors	Servo drive	DC motors
Navigation Strategy	Random bounce + timed turns	Zigzag / lawnmower pattern	Spiral pattern	S-shaped lawnmower pattern
Debris Pickup Mechanism	Vacuum fan + nozzle	Brush-only	Brush + vacuum	Vacuum fan + nozzle
Debris Storage	Removable container	Fixed bin (open top)	Mesh bag / pouch	Removable container
Motor Control	H-bridge motor driver	MOSFET driver	Direct drive (not recommended)	H-bridge motor driver
Power Source	Single 12V battery pack	7.4V Li-ion battery	9V battery for fan + 12V battery for control/drive	9V battery for vacuum fan + 12V battery for control/drive
Completion Signaling	LED + buzzer	LED only	Buzzer only	LED

Chassis / Structure	Laser-cut wood base	3D printed chassis	Foam board / acrylic	Wood base
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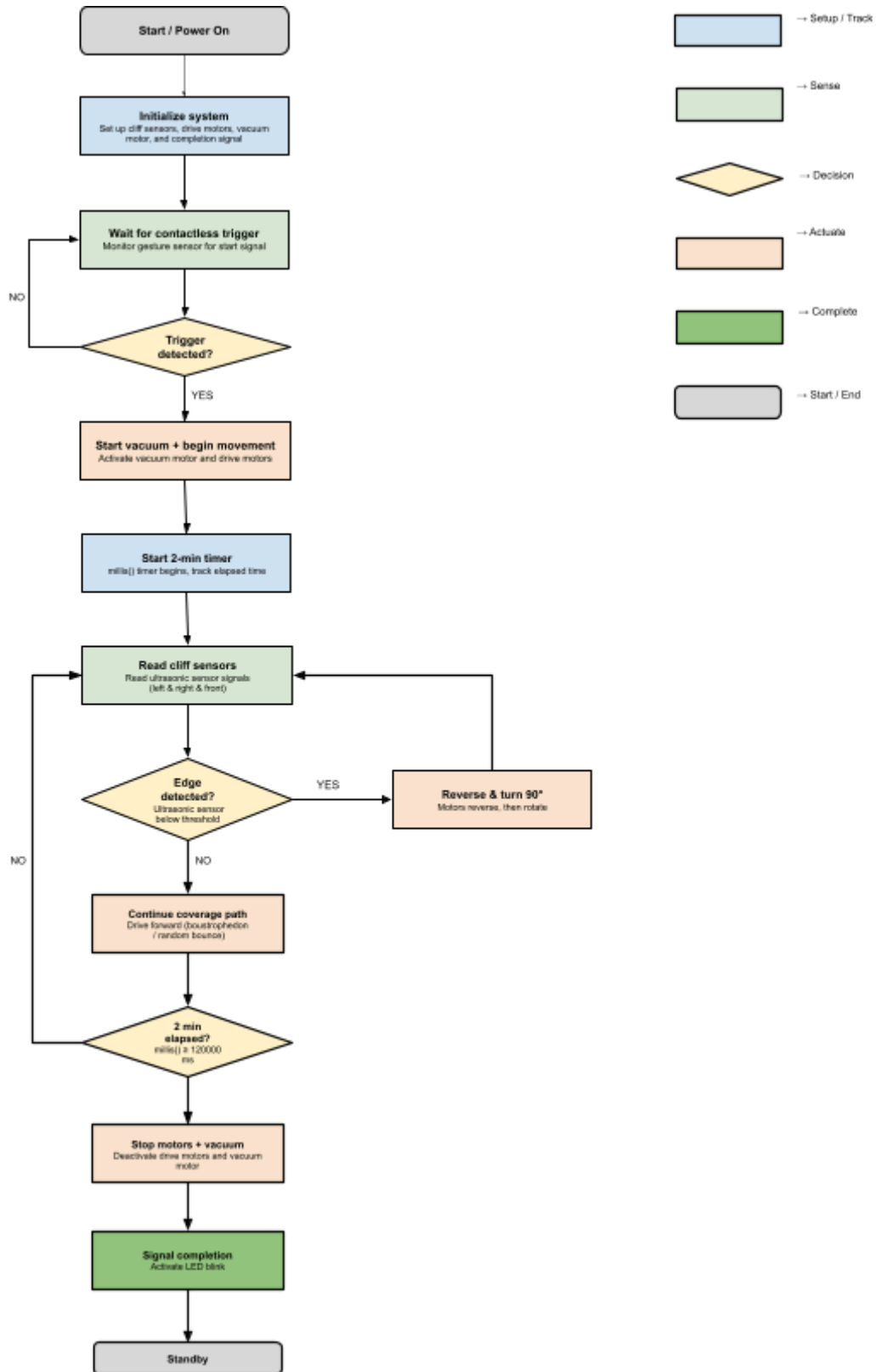
Appendix D — Bill of Materials

Part / Component	Quantity	Unit Cost(\$)	Total Cost(\$)	Source / Vendor
Gikfun Type 130 Miniature DC Motor	2	1.61	3.22	Amazon
Adafruit APDS9960 Gesture Sensor	1	7.5	7.5	Adafruit
HC-SR04 Compatible Ultrasonic Distance Sensor	3	3.95	11.85	Adafruit
Arduino Uno	1	14.99	14.99	Stockroom
Caster Wheel	1	8.99	8.99	Stockroom
H-Bridge Motor Driver Control	1	9.99	9.99	Amazon
Mesh Filter Screen	1	7.49	7.49	Amazon
12V Battery Pack	1	15.99	15.99	Stockroom
DC Fan	1	22.75	22.75	Stockroom
Allmax 9V Alkaline Battery	1	1.65	1.65	Amazon
Total Estimated Cost			104.42	

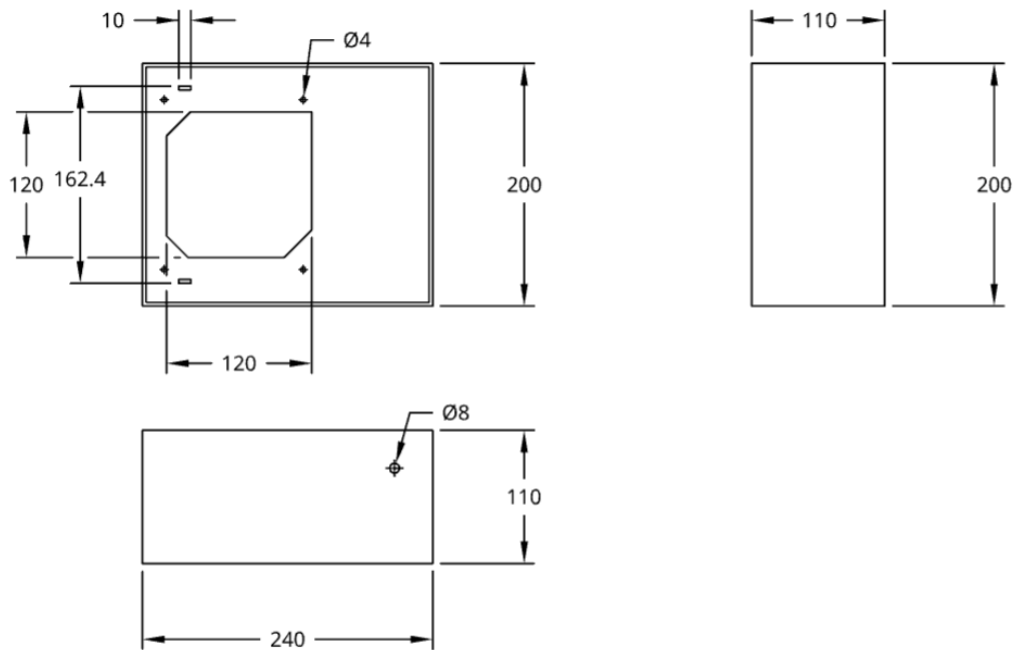
Appendix E — Electronics sketch



Appendix F — Code flow chart



Appendix G — CAD or hand drawings



Appendix H — Power budget calculations

12 V Control and Drive System					
Component	Voltage (V)	Current (mA)	Power (mW)	Quantity	Total Power (mW)
Arduino Uno	5	50	250	1	250
Gikfun DC Motor	6	400	2400	2	4800
HC-SR04 Ultrasonic Sensor	5	2.2	11	3	33
APDS9960 Gesture Sensor	3.3	1	3.3	1	3.3
LED	3.3	20	66	1	66
Total					~5152 mW

Estimated runtime:

$$12\text{ V} \times 1200\text{ mAh} = 14400\text{ mWh}$$

$$14400\text{ mWh} \div 5152\text{ mW} \approx 167\text{ minutes}$$

9 V Vacuum System					
Component	Voltage (V)	Current (mA)	Power (mW)	Quantity	Total Power (mW)
DC Fan Motor	9	200	1800	1	1800
Total					~1800 mW

Estimated runtime:

$$9\text{ V} \times 720\text{ mAh} = 6480\text{ mWh}$$

$$6480\text{ mWh} \div 1800\text{ mW} \approx 216\text{ minutes}$$

These estimated runtimes are greater than the required 5-minute operating time. Actual runtime may be lower because of motor startup current, voltage drop, and power losses in the circuit.

Appendix I — Plots and tables related to your testing.

Figure 1: Trial passing rate

